
NFDI-MatWerk/IUC02 Data schema for creep data of Ni-based superalloys including a comprehensive documentation of test results and metadata

Version 1.1 (September 2024)

This version replaces the version v1.0 (June 2024).

The main changes compared to v1.0 are listed in the last page of this document.

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Creep testing of metallic materials for high temperature applications, e.g., in turbines and power plants, yields valuable datasets. High experimental efforts are necessary to ensure stable high temperature and constant loading conditions in these long-running tests. Proper, comprehensive documentation of these experiments is a key element to enable the assessment of the quality of respective creep datasets and their targeted use (and future re-use) for specific applications.

The attached data schema for creep tests was developed within the German NFDI-MatWerk initiative (<https://nfdi-matwerk.de/>). It is intended to define a structured approach for collecting all required information on a creep experiment using the established terminology of the respective test standard ISO 204:2022. The development goals encompass the following primary aspects:

- To ensure a comprehensive description with a hierarchical data structure that can be implemented to data management platforms,
- To define a scope of documentation that allows the assessment of a dataset's quality by different end users who retrieve datasets from multiple data providers,
- To foster the exchange of high-quality creep datasets according to the FAIR principles [1], by providing easy interoperability and full reusability.

Although originally developed for datasets of Ni-based high temperature alloys, the data schema is largely agnostic to the type of material and may be similarly used for creep tests of different metallic (and other) materials.

A very detailed approach was chosen to ensure the collection of all related pieces of information, e.g. including a complete description of the material's manufacturing history and a comprehensive description of the laboratory equipment. In this way, the developed data schema serves to describe or identify high quality datasets, which can be considered as reference data for verification purposes based on their precision and documentation. It is acknowledged that certain datasets, depending on their origin and intended use, may not require this full depth of documentation. Still, the suggested data schema can help to decide which parts of information are relevant for the respective purpose. The authors intend to further optimize the data structure in future versions, defining quality classes for creep datasets based on the extent of available (meta-) information.

This version of the data schema covers creep tests with the following features:

- It considers single- and polycrystalline specimen material
- Terminology is aligned with ISO 204:2022
- Creep test under tension and constant force
- Temperature measurement with thermocouples
- Contacting extensometer system

The data schema is provided in the following formats: PDF, XLSX, CSV and JSON, and it is also available in a git repository (<https://git.rwth-aachen.de/nfdi-matwerk/iuc02>).

The XLSX/PDF/CSV version of the data schema contains 12 columns. Columns B to E are categories that define the overarching structure of the schema (see **Figure 1**). This structure mimics the way a domain expert would structure the data. Columns F to H refer to the single entries with the respective symbol and unit, if applicable. The units are community-common units. Column I is the data type. It should give a first hint on how to answer each entry field. Column J provides exemplary answers for most of the entries and the options of the corresponding drop-down list, if applicable. Column K is the requirement profile, while column L includes further explanations on the requirement and on the applicability of some entries.

The Requirement Profile (column K) refers to the highest quality class of reference creep data, taken from calibrated instruments, and which shall enable the following usages:

1. Checking own creep test results on nominally similar material
2. Verification of own testing set-up (e.g. by testing same/similar material)
3. Using the data as input data for simulations of creep behavior for design and alloy development

Other quality classes or typical research datasets require less documentation.

The JSON schema yields the same categorization as the XLSX/PDF/CSV files (cf. Figure 1) and includes those concepts present in the data schema. The requirement profile and data type are essential features to be provided for each entry. The JSON schema is structured with column B as the first level of categorization and the mapping further builds upon it.

Our current definition for reference data of materials is available [here](#) [3].

Metadata	Test info	Test job details	
		Test parameters	
		Related research outcome	
	Material related	History and condition of the material	As-manufactured material
			As-tested material
			Heat treatment
			Chemical composition
			Microstructure
			Results from NDT
			Results from other mechanical tests
			...
		Test piece	
	Measuring and test equipment	Test machine	Data acquisition (e.g., unit)
			...
		Test force	Loading system
			Load sensor
			Data acquisition (e.g., frequency)
		Laboratory conditions	
		Temperature-measuring system	Measuring instrument
			Data acquisition (e.g., calibrated range)
			...
		Extensometer system	Contacting extensometer
			Non-contacting extensometer
		...	
		Data processing procedures	
Primary data	Test result	Values recorded at test start	
		Values recorded during test run	
		Values recorded after end of test	
Secondary data	Test result	Values recorded during test run	
		Elongation values	
		Extension values	

Figure 1. Data schema structure for creep data of Ni-based superalloys. The shown excerpt of the full data schema illustrates the four overarching levels of categorization, corresponding to columns B to E in the attached files.

Changes compared to v1.0

The amendments were mainly editorial and formal changes and are as follows:

- Column "Comment": Text in row 193 was revised and cross-references to rows were revised and corrected for all entries.
- Row 142, column "Requirement": "Deviation detected during calibration" (data acquisition temperature-measuring system) changed to optional.
- Rows 164 and 165, column "Entry": "...primary data and processed data" was changed to "...primary and processed data series".
- Column "Exemplary answer / options (in case of drop-down list)": Capitalization was revised and corrected.

The above-mentioned changes refer to the XLSX and CSV files. The JSON schema was adapted accordingly. The data schema structure, **Figure 1**, remained unchanged.

References

[1] <https://nfdi-matwerk.de/>

[2] Wilkinson, M., Dumontier, M., Aalbersberg, I. et al. The FAIR Guiding Principles for scientific data management and stewardship. *Sci Data* 3, 160018 (2016). <https://doi.org/10.1038/sdata.2016.18>

[3] Hickel, T., Richter, S., Bitzek, E., Ávila Calderón, L. A., Gedsun, A., Forti, M., Hammerschmidt, T., Olbricht, J., & Skrotzki, B. (2024). NFDI-MatWerk/IUC02 Definition for Reference Data of Materials (1.0). Zenodo. <https://doi.org/10.5281/zenodo.11667673>

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